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MATH 455: Partial Differential Equations

Assignment #3 - Wave Equation
Due Tuesday, October 6, 2026

1. (15 points) Define the functions

$$f(x) = \begin{cases} x, & \text{if } 0 \le x \le 1 \\ 1, & \text{if } 1 < x < 5 \\ 6 - x, & \text{if } 5 \le x \le 6 \\ 0, & \text{else,} \end{cases}$$

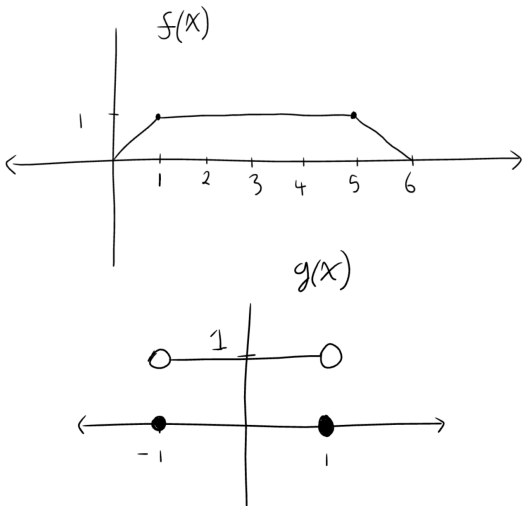
$$g(x) = \begin{cases} 1, & \text{if } |x| < 1 \\ 0, & \text{else,} \end{cases}$$

and consider solutions of the wave equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0. \tag{WE}$$

(a) Graph $f(x)$ and $g(x)$ by hand.

Solution —



(b) Find the solution of (WE) for $x \in \mathbb{R}, t > 0$ with

$$u(0,x) = f(x) \quad \text{and} \quad \frac{\partial u}{\partial t}(0,x) = 0$$

for $x \in \mathbb{R}$. In particular, write the solution in terms of the $f(x)$ function. Then, graph by hand $u(\frac{1}{c}, x)$, $u(\frac{2}{c}, x)$, and $u(\frac{3}{c}, x)$ using different axes.

Solution — To find the general solution of (WE), apply the substitution

$$r(t,x) = x - ct$$
$$s(t,x) = x + ct$$

Assume $u(t,x) = U(r,s)$ is a solution and substitute (we first calculate the needed derivatives of u):

$$\frac{\partial u}{\partial x} = \frac{\partial U}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial U}{\partial s}\frac{\partial s}{\partial x}$$

$$= \frac{\partial U}{\partial r} + \frac{\partial U}{\partial s}$$

And

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial U}{\partial r} + \frac{\partial U}{\partial s}\right)$$

To simplify further, note that $\frac{\partial U}{\partial r}$ is some function of r and s, say $\phi_1(r,s)$, and $\frac{\partial U}{\partial s}$ is also some function of r and s, say $\phi_2(r,s)$:

$$= \frac{\partial}{\partial x}\left(\phi_1(r,s) + \phi_2(r,s)\right)$$

$$= \frac{\partial \phi_1}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial \phi_1}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial \phi_2}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial \phi_2}{\partial s}$$

$$= \frac{\partial \phi_1}{\partial r} + \frac{\partial \phi_1}{\partial s} + \frac{\partial \phi_2}{\partial r} + \frac{\partial \phi_2}{\partial s}$$

$$= \frac{\partial}{\partial r}\left(\frac{\partial U}{\partial r}\right) + \frac{\partial}{\partial s}\left(\frac{\partial U}{\partial r}\right) + \frac{\partial}{\partial r}\left(\frac{\partial U}{\partial s}\right) + \frac{\partial}{\partial s}\left(\frac{\partial U}{\partial s}\right)$$

$$= \frac{\partial^2 U}{\partial r^2} + \frac{\partial^2 U}{\partial s \partial r} + \frac{\partial^2 U}{\partial r \partial s} + \frac{\partial^2 U}{\partial s^2}$$

Let us assume that $U(r,s)$ is smooth and well-behaved enough for the cross-derivatives to be equal. Then

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 U}{\partial r^2} + 2\left(\frac{\partial^2 U}{\partial r \partial s}\right) + \frac{\partial^2 U}{\partial s^2}$$

By similar mechanics, we obtain

$$\frac{\partial^2 u}{\partial t^2} = c^2\left(\frac{\partial^2 U}{\partial r^2} - 2\left(\frac{\partial^2 U}{\partial r \partial s}\right) + \frac{\partial^2 U}{\partial s^2}\right)$$

Substituting these second derivatives in the general wave equation yields:

$$\left[c^2\left(\frac{\partial^2 U}{\partial r^2} - 2\left(\frac{\partial^2 U}{\partial r \partial s}\right) + \frac{\partial^2 U}{\partial s^2}\right)\right]$$
$$-c^2\left[\frac{\partial^2 U}{\partial r^2} + 2\left(\frac{\partial^2 U}{\partial r \partial s}\right) + \frac{\partial^2 U}{\partial s^2}\right]$$

$$= -4c^2\left(\frac{\partial^2 U}{\partial r \partial s}\right) = 0$$

$$\iff \frac{\partial^2 U}{\partial r \partial s} = 0$$

Our change-of-variables has transformed the original wave equation into this more workable form, which we can now simply integrate to extract U:

$$\frac{\partial}{\partial r}\frac{\partial U}{\partial s} = 0$$

$$\iff \frac{\partial U}{\partial s} = C_1(s)$$

$$\iff U = C_2(s) + C_3(r)$$

$$\iff U(r,s) = F(r) + G(s)$$

$$\iff u(t,x) = F(x-ct) + G(x+ct)$$

We now apply our initial conditions.

$$u(0,x) = f(x) \implies F(x) + G(x) = f(x)$$

$$\left.\frac{\partial u}{\partial t}\right|_{(0,x)} = 0 \implies -cF'(x) + cG'(x) = 0$$
$$\implies F'(x) = G'(x)$$
$$\implies F(x) = G(x) + \delta \quad \text{for some } \delta \in \mathbb{R}$$

Our two constraints combine to imply

$$2G(x) + \delta = f(x)$$
$$\implies G(x) = \frac{1}{2}(f(x) - \delta)$$
$$\implies F(x) = \frac{1}{2}(f(x) + \delta)$$

Hence

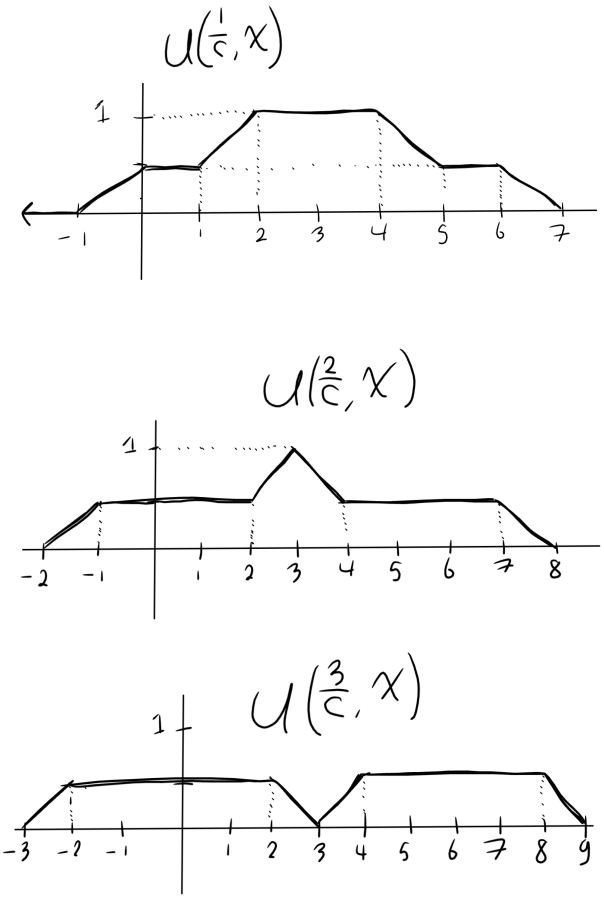
$$u(t,x) = \frac{1}{2}(f(x-ct) + \delta) + \frac{1}{2}(f(x+ct) - \delta)$$

$$= \boxed{\frac{1}{2}f(x-ct) + \frac{1}{2}f(x+ct)}$$

$$\boxed{\begin{aligned} u\left(\frac{1}{c}, x\right) &= \frac{1}{2}f(x-1) + \frac{1}{2}f(x+1) \\ u\left(\frac{2}{c}, x\right) &= \frac{1}{2}f(x-2) + \frac{1}{2}f(x+2) \\ u\left(\frac{3}{c}, x\right) &= \frac{1}{2}f(x-3) + \frac{1}{2}f(x+3) \end{aligned}}$$

(c) Find the solution of (WE) for $x \in \mathbb{R}, t > 0$ with

$$u(0,x) = 0 \quad \text{and} \quad \frac{\partial u}{\partial t}(0,x) = g(x)$$



for $x \in \mathbb{R}$. In particular, write the solution in terms of the $g(x)$ function. Then, graph by hand $u(\frac{1}{c}, x)$, $u(\frac{2}{c}, x)$, and $u(\frac{3}{c}, x)$ using different axes.

Solution — Now we apply different initial conditions to the same general solution we obtained $u(t, x) = F(x - ct) + G(x + ct)$:

$$u(0, x) = 0 \implies F(x) = -G(x)$$

$$\left.\frac{\partial u}{\partial t}\right|_{(0,x)} = g(x) \implies -cF'(x) + cG'(x) = g(x)$$

From our first constraint, $F'(x) = -G'(x)$, so

$$-c(-G'(x)) + cG'(x) = g(x)$$

$$G'(x) = \frac{g(x)}{2c}$$

$$G(x) = \frac{1}{2c}\int g(x)dx + \delta$$

$$F(x) = -\frac{1}{2c}\int g(x)dx - \delta$$

Let $A(x)$ be the antiderivative of g, so that $A'(x) = g(x)$:

$$G(x) = \frac{1}{2c}A(x) + \delta \qquad F(x) = -\frac{1}{2c}A(x) - \delta$$

Hence

$$u(t,x) = \left(-\frac{1}{2c}A(x-ct) - \delta\right) + \left(\frac{1}{2c}A(x+ct) + \delta\right)$$

$$= \frac{1}{2c}\left(A(x+ct) - A(x-ct)\right)$$

$$= \boxed{\frac{1}{2c}\int_{x-ct}^{x+ct} g(\zeta)d\zeta}$$

We compute a few solution functions at various times. First, note that the area under g from a to b is

$$A(x) = \begin{cases} 0, & \text{if } x < -1 \\ x+1, & \text{if } |x| < 1 \\ 2, & \text{if } x > 1 \end{cases}$$

Now, at different time increments $\frac{n}{c}$ for $n = 1, 2, 3 \cdots$, the solution is

$$u\left(\frac{n}{c}, x\right) = \frac{1}{2c}\int_{x-n}^{x+n} g(\zeta)d\zeta$$

$$= \frac{1}{2c}\Big(A(x+n) - A(x-n)\Big)$$

$$= \frac{1}{2c}\left(\begin{cases} 0, & \text{if } x+n < -1 \\ (x+n)+1, & \text{if } |x+n| < 1 \\ 2, & \text{if } x+n > 1 \end{cases} - \begin{cases} 0, & \text{if } x-n < -1 \\ (x-n)+1, & \text{if } |x-n| < 1 \\ 2, & \text{if } x-n > 1 \end{cases}\right)$$

$$= \frac{1}{2c}\begin{cases} 0 & \text{if } x \le -n-1 \\ n+1+x, & \text{if } -n-1 \le x \le -n+1 \\ 2, & \text{if } -n+1 \le x < n-1 \\ n+1-x, & \text{if } n-1 \le x \le n+1 \\ 0, & \text{if } x \ge n+1 \end{cases}$$

Consider the symmetry of the $n+1+x$ and $n+1-x$ cases. If we apply the substitution $A = n-1$ and $B = n+1$, then these cases reduce to

$$\begin{cases} B+x & \text{if } -B \le x \le -A \\ B-x & \text{if } A \le x \le B \end{cases}$$

Which reduces to

$$\Big\{B - |x| \quad \text{if } A \le |x| \le B$$

(Note that $B+x = B-|x|$ when x is negative (in the first interval) and $B-x = B-|x|$ when x is positive (in the second interval)). Also note that the two zero cases can be described by $|x| \ge n+1$. Hence

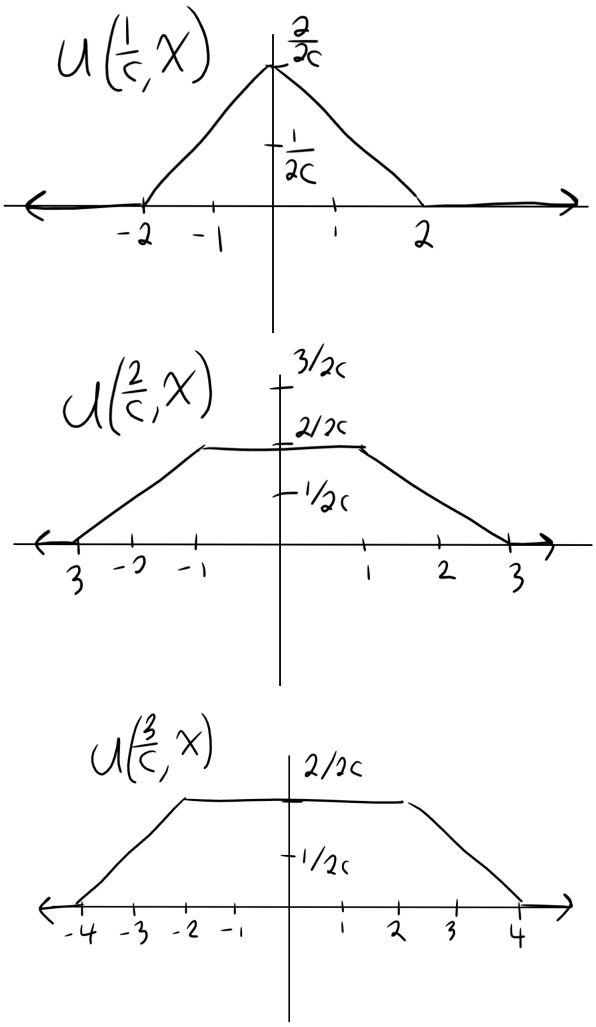
$$u\left(\frac{n}{c}, x\right) = \frac{1}{2c}\begin{cases} 0 & \text{if } |x| \ge n+1 \\ n+1-|x| & \text{if } n-1 \le |x| \le n+1 \\ 2 & \text{if } |x| \le n-1 \end{cases}$$

Hence

$$\boxed{u\left(\frac{1}{c}, x\right) = \frac{1}{2c}\begin{cases} 0 & \text{if } |x| \ge 2 \\ 2-|x| & \text{if } 0 \le |x| \le 2 \end{cases}}$$

$$\boxed{u\left(\frac{2}{c}, x\right) = \frac{1}{2c}\begin{cases} 0 & \text{if } |x| \ge 3 \\ 3-|x| & \text{if } 1 \le |x| \le 3 \\ 2 & \text{if } |x| \le 1 \end{cases}}$$

$$u\left(\frac{3}{c},x\right)=\frac{1}{2c}\begin{cases}0 & \text{if } |x|\geq 4\\ 4-|x| & \text{if } 2\leq|x|\leq 4\\ 2 & \text{if } |x|\leq 2\end{cases}$$



2. (12 points) Let $f, g \in \mathrm{PC}^1[0, L]$ be given continuous functions. Find the solution of (WE) for $0 < x < L$ and $t > 0$ with the boundary conditions

$$\frac{\partial u}{\partial x}(t,0)=\frac{\partial u}{\partial x}(t,L)=0, \qquad t>0$$

and the initial conditions for $0 < x < L$

$$u(0,x)=f(x), \quad \frac{\partial u}{\partial t}(0,x)=g(x).$$

Solution — Since we now impose boundary conditions (in addition to initial conditions), we must now solve the wave equation on a bounded domain, which requires separation of variables. We assume the solution takes the separated form

$$u(t,x) = T(t)X(x)$$

For all $t, x \in \mathbb{R}$. Then

$$\frac{\partial^2 u}{\partial t^2} = T''(t)X(x) \qquad \frac{\partial^2 u}{\partial x^2} = T(t)X''(x)$$

and the wave equation takes the form

$$T''(t)X(x) - c^2 T(t)X''(x) = 0 \implies \frac{T''(t)}{T(t)} = c^2 \frac{X''(x)}{X(x)}$$

which is of the form $F(t) = G(x)$. Because we're assuming this equality holds for all x, t, then both sides of the equation must be constant (fix $x = x_0$ and we have $F(t) = G(x_0)$, which is equivalent to saying that $F(t)$ is (in general) equal to some constant $G(x_0)$. The same can be deduced by fixing $t = t_0$). Let λ be this constant:

$$\frac{T''(t)}{T(t)} = c^2 \frac{X''(x)}{X(x)} = \lambda \implies \begin{cases} T'' - \lambda T = 0 \\ c^2 X'' - \lambda X = 0 \end{cases}$$

This is a system of independent second-order homogeneous linear ODEs, each of which can be solved with the characteristic equation to obtain the solutions

$$T(t) = c_1 e^{-\sqrt{\lambda}t} + c_2 e^{\sqrt{\lambda}t}$$
$$X(x) = c_2 e^{-\frac{\sqrt{\lambda}}{c}x} + c_3 e^{\frac{\sqrt{\lambda}}{c}x}$$

When $\lambda > 0$, or

$$T(t) = A + Bt$$
$$X(x) = C + Dx$$

When $\lambda = 0$, or

$$T(t) = A_1 \cos(\sqrt{-\lambda}t) + B_1 \sin(\sqrt{-\lambda}t)$$
$$X(x) = C_1 \cos\left(\frac{\sqrt{-\lambda}}{c}x\right) + D_1 \sin\left(\frac{\sqrt{-\lambda}}{c}x\right)$$

When $\lambda < 0$. We will now apply our boundary conditions

$$\frac{\partial u}{\partial x}(t,0) = \frac{\partial u}{\partial x}(t,L) = 0$$

Note that these are Neumann boundary conditions because they fix the physical slope of the string in space at the boundaries (as opposed to fixing the position of the string in space with Dirichlet boundaries).

$$\frac{\partial u}{\partial x}(t,0) = T(t)X'(x)\mid_{(t,0)} = T(t)X'(0) = 0$$
$$\frac{\partial u}{\partial x}(t,L) = T(t)X'(x)\mid_{(t,L)} = T(t)X'(L) = 0$$

For $\lambda > 0$: Ignoring the trivial case where $T(t) = 0$ (which causes the wave solution $u(t,x) = T(t)X(x)$ to be zero everywhere), these initial conditions reduce to

$$\begin{aligned} & X'(0) = 0 && X'(L) = 0 \\ \implies\ & c_2 = c_3 && \implies -c_2 e^{\frac{-\sqrt{\lambda}}{c}L} + c_3 e^{\frac{\sqrt{\lambda}}{c}L} = 0 \\ & && \implies c_3 \left[2\sinh \frac{\sqrt{\lambda}}{c} L \right] = 0 \\ & && \implies c_3 = 0 \quad (\text{because } \lambda \neq 0,\ c \neq 0,\ L \neq 0) \\ & && \implies c_2 = 0 \end{aligned}$$

$$\implies X(x) = 0 \implies u(t,x) = 0$$

For $\lambda = 0$:

$$\begin{aligned} & X'(0) = 0 && X'(L) = 0 \\ \implies\ & D = 0 && \implies D = 0 \\ & && \implies X(x) = C \\ & && \implies u(t,x) = C(A + Bt) \end{aligned}$$

For $\lambda < 0$:

$$\begin{aligned} X'(0) = 0 \qquad & X'(L) = 0 \\ \implies \frac{D_1\sqrt{-\lambda}}{c} = 0 \qquad & \implies -C_1 \sin\left(\frac{\sqrt{-\lambda}}{c}L\right) + D_1 \cos\left(\frac{\sqrt{-\lambda}}{c}L\right) = 0 \\ \implies D_1 = 0 \quad (\text{since } \lambda \neq 0,\, c \neq 0) \qquad & \implies C_1 \sin\left(\frac{\sqrt{-\lambda}}{c}L\right) = 0 \quad (\text{substituting } D_1 = 0) \\ & \implies \frac{\sqrt{-\lambda}}{c}L = n\pi \quad (\text{For nontrivial } C_1 \neq 0) \\ & \implies \lambda_n = -\left(\frac{n\pi c}{L}\right)^2 \end{aligned}$$

Hence

$$T(t) = A_1 \cos\left(\frac{n\pi c}{L}t\right) + B_1 \sin\left(\frac{n\pi c}{L}t\right)$$
$$X(x) = C_1 \cos\left(\frac{n\pi x}{L}\right)$$

and

$$u(t,x) = \left[A_1 \cos\left(\frac{n\pi c}{L}t\right) + B_1 \sin\left(\frac{n\pi c}{L}t\right)\right] C_1 \cos\left(\frac{n\pi x}{L}\right)$$

By linearity of the wave equation, the superposition principle states that the general solution is the sum of all separated solution modes (for each λ). Hence, the provided Neumann boundary conditions lead to the following general solution (over all possible λ)

$$\boxed{u(t,x) = a_0 + b_0 t + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi c}{L}t\right) + b_n \sin\left(\frac{n\pi c}{L}t\right)\right] \cos\left(\frac{n\pi x}{L}\right)}$$

We now apply the initial conditions:

$$u(0,x) = f(x) \implies a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) = f(x)$$

$$\left.\frac{\partial u}{\partial t}\right|_{(0,x)} = g(x) \implies b_0 + \left(\frac{\pi c}{L}\right)\sum_{n=1}^{\infty} n b_n \cos\left(\frac{n\pi x}{L}\right) = g(x)$$

Hence the a_n coefficients must be exactly the cosine-series coefficients of the Fourier expansion of $f(x)$ on $[0, L]$. Likewise, the the velocity condition, the product $(n\pi c/L)b_n$ must be exactly the cosine-series coefficients of the Fourier expansion of $g(x)$ on $[0, L]$. Note that these two series must exist due to $f, g \in PC^1[0, L]$

$$\boxed{a_0 = \frac{1}{L}\int_0^L f(x)dx}$$

$$\boxed{a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)dx}$$

$$\boxed{b_0 = \frac{1}{L}\int_0^L g(x)dx}$$

$$\frac{n\pi c}{L}b_n = \frac{2}{L}\int_0^L g(x)\cos\left(\frac{n\pi x}{L}\right)dx \implies \boxed{b_n = \frac{2}{n\pi c}\int_0^L g(x)\cos\left(\frac{n\pi x}{L}\right)dx}$$

3. (15 points) Assume $u(t,x)$ satisfies (WE) for $0 < x < L$ and $t > 0$ and define

$$\mathcal{E}(t) = \frac{1}{2}\int_0^L \left|\frac{\partial u}{\partial t}(t,x)\right|^2 dx + \frac{1}{2}c^2\int_0^L \left|\frac{\partial u}{\partial x}(t,x)\right|^2 dx$$

to be the associated total energy, where the first term in $\mathcal{E}(t)$ represents the kinetic energy and the second represents the potential energy.

(a) Use **integration by parts** to show that the total energy satisfies

$$\mathcal{E}'(t) = c^2\frac{\partial u}{\partial t}(t,x)\frac{\partial u}{\partial x}(t,x)\Big|_{x=0}^{x=L}.$$

Solution — Note that $|A|^2 = A^2$ regardless of the sign of A. Hence the absolute values in $\mathcal{E}(t)$ can be dropped:

$$\mathcal{E}(t) = \frac{1}{2}\int_0^L \left(\frac{\partial u}{\partial t}\right)^2 dx + \frac{1}{2}c^2\int_0^L \left(\frac{\partial u}{\partial x}\right)^2 dx$$

Then

$$\mathcal{E}'(t) = \frac{d}{dt}\left[\frac{1}{2}\int_0^L \left(\frac{\partial u}{\partial t}\right)^2 dx + \frac{1}{2}c^2\int_0^L \left(\frac{\partial u}{\partial x}\right)^2 dx\right]$$

$$= \frac{1}{2}\int_0^L \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial t}\right)^2 dx + \frac{1}{2}c^2\int_0^L \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x}\right)^2 dx$$

$$= \int_0^L \left(\frac{\partial u}{\partial t}\right)\underbrace{\left(\frac{\partial^2 u}{\partial t^2}\right)}_{c^2\frac{\partial^2 u}{\partial x^2}} dx + c^2\int_0^L \left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial^2 u}{\partial t\partial x}\right)dx$$

since $u(t,x)$ satisfies the wave equation

$$= c^2 \int_0^L \left(\frac{\partial u}{\partial t}\right)\left(\frac{\partial^2 u}{\partial x^2}\right) dx + c^2 \int_0^L \left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial^2 u}{\partial t \partial x}\right) dx$$

We can now apply integration by parts to the first term. Let

$$\mathbf{U} = \frac{\partial u}{\partial t}$$

Note that the partial derivative of a function of independent parameters (parameters which themselves aren't functions of other variables) is equivalent to the total derivative because both types of derivatives simply treat everything else as a constant in this context. Hence, we can write

$$\frac{dU}{dx} = \frac{d}{dx}U = \frac{\partial}{\partial x}U = \frac{\partial^2 u}{\partial x \partial t} \implies \mathbf{dU} = \frac{\partial^2 u}{\partial x \partial t} dx$$

Let

$$\mathbf{dV} = \left(\frac{\partial^2 u}{\partial x^2}\right) dx \implies \mathbf{V} = \int \left(\frac{\partial^2 u}{\partial x^2}\right) dx$$

$$= \frac{\partial u}{\partial x} + C(t) = \frac{\partial u}{\partial x}$$

Note that we can ignore the "constant" of integration $C(t)$ because it always cancels through the process of integration by parts.

Hence

$$\mathcal{E}'(t) = c^2 \left[\left(\frac{\partial u}{\partial t}\right)\left(\frac{\partial u}{\partial x}\right)\Big|_{x=0}^{x=L} - \int_0^L \left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial^2 u}{\partial x \partial t}\right) dx \right] + c^2 \int_0^L \left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial^2 u}{\partial t \partial x}\right) dx$$

Assuming $u(t,x)$ is sufficiently smooth, the cross derivatives are equal and their corresponding terms will thus cancel out, giving:

$$\mathcal{E}'(t) = c^2 \left(\frac{\partial u}{\partial t}\right)\left(\frac{\partial u}{\partial x}\right)\Big|_{x=0}^{x=L}$$

(b) Show that the energy is conserved, meaning $\mathcal{E}(t) = \mathcal{E}(0)$ for all $t \geq 0$, for the zero Neumann boundary condition, namely assume that for $t \geq 0$

$$\frac{\partial u}{\partial x}(t,0) = 0 \qquad \text{and} \qquad \frac{\partial u}{\partial x}(t,L) = 0.$$

Solution —

$$\mathcal{E}'(t) = c^2 \left(\frac{\partial u}{\partial t}\right)\left(\frac{\partial u}{\partial x}\right)\Bigg|_{x=0}^{x=L}$$

$$= c^2 \left[\frac{\partial u}{\partial t}(t,L)\underbrace{\frac{\partial u}{\partial x}(t,L)}_{0} - \frac{\partial u}{\partial t}(t,0)\underbrace{\frac{\partial u}{\partial x}(t,0)}_{0}\right] = 0$$

And $\mathcal{E}'(t) = 0 \implies \mathcal{E}(0) = \mathcal{E}(t)$ for all t.

(c) Show that the energy is conserved for any Dirichlet boundary conditions, namely for given $A, B \in \mathbb{R}$ assume that for $t \geq 0$

$$u(t,0) = A \qquad \text{and} \qquad u(t,L) = B.$$

Solution —

$$u(t,0) = A \implies \frac{\partial u}{\partial t}(t,0) = 0$$

$$u(t,L) = B \implies \frac{\partial u}{\partial t}(t,L) = 0$$

And

$$\mathcal{E}'(t) = c^2 \left[\underbrace{\frac{\partial u}{\partial t}(t,L)}_{0}\frac{\partial u}{\partial x}(t,L) - \underbrace{\frac{\partial u}{\partial t}(t,0)}_{0}\frac{\partial u}{\partial x}(t,0)\right] = 0$$

Hence energy is conserved.

(d) Show that the potential energy and kinetic energy are equal for a traveling wave solution

$$u(t,x) = v(x-ct)$$

where v is a sufficiently smooth function.

Solution — Take the kinetic and potential energy of a wave function $u(t,x)$ to be, respectively,

$$K = \frac{1}{2}\int_0^L \left|\frac{\partial u}{\partial t}\right|^2 dx \qquad U = \frac{1}{2}c^2\int_0^L \left|\frac{\partial u}{\partial x}\right|^2 dx$$

For $u(t,x) = v(x-ct)$,

$$K = \frac{1}{2}\int_0^L \left|\frac{\partial}{\partial t}v(x-ct)\right|^2 dx = \frac{1}{2}\int_0^L |(-c)v'(x-ct)|^2\, dx$$

$$= \frac{1}{2}c^2\int_0^L (v'(x-ct))^2 dx$$

And

$$U = \frac{1}{2}c^2\int_0^L \left|\frac{\partial}{\partial x}v(x-ct)\right|^2 dx = \frac{1}{2}c^2\int_0^L (v'(x-ct))^2 dx$$

$$= \frac{1}{2}c^2\int_0^L |v'(x-ct)|^2\, dx$$

Hence $K = U$ for such a wave.

Note that we only need v to be smooth enough to have a first derivative (the only derivative of v this proof depends on existing).

4. (8 points) Maxwell's equations for electric $E(t,x)$ and magnetic $B(t,x)$ fields in a vacuum are

$$\begin{cases} \dfrac{\partial B}{\partial t} + \nabla \times E = 0 \\ \mu_0\epsilon_0\dfrac{\partial E}{\partial t} - \nabla \times B = 0 \\ \nabla \cdot E = \nabla \cdot B = 0 \end{cases}$$

where $\mu_0 > 0$ is the permeability of free space and $\epsilon_0 > 0$ is the permittivity of free space. Using the "curl-curl" identity

$$\nabla \times (\nabla \times \phi) = \nabla (\nabla \cdot \phi) - \Delta \phi$$

for any $\phi(x)$, show that both E and B satisfy the (vector) wave equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \Delta u = 0$$

and determine the associated speed of propagation c. Here, $\Delta \phi = \nabla \cdot (\nabla \phi)$ is the Laplacian of ϕ, which represents a generalization of the second derivative in higher dimensions.

Solution — This problem reduces to showing that if a vector function satisfies Maxwell's equations, then it must necessarily satisfy the wave equation.

Note that the presence of $\nabla \times E$ and $\nabla \times B$ in the equations indicates that E and B must be vector quantities. Hence we will notate them as $\mathbf{E}$ and $\mathbf{B}$. A vacuum is also 3-dimensional which means the spatial parameter x in $\mathbf{E}(t, x)$ and $\mathbf{B}(t, x)$ is actually a 3-dimensional vector itself. So the unambiguous notation here would be $\mathbf{E}(t, \mathbf{x})$ and $\mathbf{B}(t, \mathbf{x})$

Proof — Let an arbitrary $\mathbf{E}(t, \mathbf{x})$ and $\mathbf{B}(t, \mathbf{x})$ satisfy Maxwell's equations. Then $\mathbf{E}$ and $\mathbf{B}$ must satisfy the second Maxwell equation:

$$\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} - \nabla \times \mathbf{B} = 0$$

Taking the time derivative of both sides, we get that

$$\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} - \frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = 0$$

Dividing through by the constant factor, we get something that is almost the wave equation for $\mathbf{E}$:

$$\frac{\partial^2 \mathbf{E}}{\partial t^2} - \frac{1}{\mu_0 \epsilon_0} \left(\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) \right) = 0 \tag{1}$$

All we need to do is show that $\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = \Delta \mathbf{E}$.

Note that it can easily be shown that the partial time derivative operator and the curl operator commute with each other if the function is smooth enough to have equal cross-derivatives. That is,

$$\frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = \nabla \times \frac{\partial \mathbf{B}}{\partial t}$$

Hence we can write equation (1) as

$$\frac{\partial^2 \mathbf{E}}{\partial t^2} - \frac{1}{\mu_0 \epsilon_0} \left(\nabla \times \frac{\partial \mathbf{B}}{\partial t} \right) = 0$$

And note that the first Maxwell equation establishes that $\partial \mathbf{B}/\partial t = -\nabla \times E$. Substituting this into the left-hand side, we get

$$\frac{\partial^2 \mathbf{E}}{\partial t^2} - \frac{1}{\mu_0 \epsilon_0} (\nabla \times (-\nabla \times \mathbf{E}))$$

$$= \frac{\partial^2 \mathbf{E}}{\partial t^2} + \frac{1}{\mu_0 \epsilon_0} (\nabla \times (\nabla \times \mathbf{E}))$$

$$= \frac{\partial^2 \mathbf{E}}{\partial t^2} + \frac{1}{\mu_0 \epsilon_0} \left(\nabla (\underbrace{\nabla \cdot \mathbf{E}}_{0}) - \Delta \mathbf{E} \right) \quad \text{(by curl-curl identity)}$$

$$= \frac{\partial^2 \mathbf{E}}{\partial t^2} - \frac{1}{\mu_0 \epsilon_0} \Delta \mathbf{E} = 0$$

Hence $\mathbf{E}$ must satisfy the wave equation with speed of propagation

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

We now show that $\mathbf{B}$ satisfies the wave equation by a similar method. From the first Maxwell equation, we have that

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0$$

$$\implies \frac{\partial^2 \mathbf{B}}{\partial t^2} + \frac{\partial}{\partial t}(\nabla \times \mathbf{E}) = 0$$

$$= \frac{\partial^2 \mathbf{B}}{\partial t^2} + \left(\nabla \times \frac{\partial \mathbf{E}}{\partial t}\right)$$

$$= \frac{\partial^2 \mathbf{B}}{\partial t^2} + \left(\nabla \times \frac{1}{\mu_0 \epsilon_0}(\nabla \times \mathbf{B})\right) \qquad \text{(by Maxwell's 2nd equation)}$$

$$= \frac{\partial^2 \mathbf{B}}{\partial t^2} + \frac{1}{\mu_0 \epsilon_0}(\nabla \times (\nabla \times \mathbf{B}))$$

$$= \frac{\partial^2 \mathbf{B}}{\partial t^2} + \frac{1}{\mu_0 \epsilon_0}\left(\nabla(\underbrace{\nabla \cdot \mathbf{B}}_{0}) - \Delta \mathbf{B}\right)$$

$$= \frac{\partial^2 \mathbf{B}}{\partial t^2} - \frac{1}{\mu_0 \epsilon_0}\Delta \mathbf{B} = 0$$

Hence $\mathbf{B}$ must satisfy the wave equation with identical speed of propagation

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$